6.1

Why do you need food?

All living things need food. Green plants make their own food using energy from the sun. Animals, however, need to eat in order to stay alive.

Food for energy

All living things — including you — need energy to grow and stay alive. You need energy in order to move. You need energy to walk, run, jump, cycle and do all of those activities that require movement. Even if you remained totally still, your body would use up energy just to keep working. Energy is needed to keep you breathing, your heart pumping and your body temperature at 37°C, even when the air temperature changes.

Food for growth and repair of cells

While you are young and still growing, new cells are being produced very rapidly. Food is needed to supply the raw materials that the new cells are made of. Even after you stop growing, new cells are needed to replace those that die. Many of your cells have quite short lifetimes and need constant replacement. Red blood cells live for only about 120 days. There are so many of them in your body that about 1.7 million new red blood cells have to be produced every second. Your outer layer of skin replaces itself about every 19 days. When cells are damaged or die as a result of illness or injury, they need to be repaired or replaced.

The chemicals in food

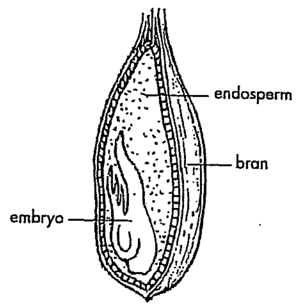
The useful and nutritious substances that make up food are called **nutrients**. Nutrients fall into five main groups:

- carbohydrates
- fats and oils
- proteins
- vitamins
- minerals.

In one end and out the other

Foods contain other important substances that are not nutrients, and are therefore not used for energy or for growth and repair. Fibre and water are two of these substances. **Fibre** is a substance found in the walls of plant cells that is only partially broken down by your digestive system. Although it really does go 'in one end and out the other', it serves a very useful purpose and is an essential part of your diet. It provides bulk to your food, allowing it to move properly through your intestines. Without fibre, undigested food travels too slowly through the large intestine, losing too much water. The result is difficulty in releasing the solid food waste from the body, a condition called **constipation**. Lack of fibre in the diet can also lead to haemorrhoids (varicose veins around the anal passage, also known as piles), bowel cancer and several other diseases.

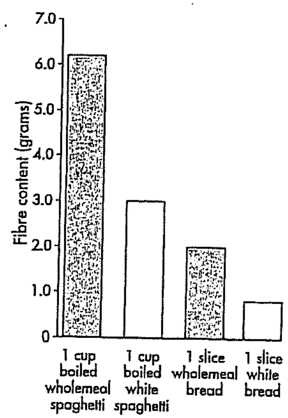
Fibre can be found only in foods that come from plants — foods such as fruits and vegetables, wholegrain breads and cereals, nuts and seeds.



A grain of wheat

Wholegrain products are higher in dietary fibre because they contain the outer covering, or bran, of the grain. When grains are highly processed, as they are in the production of white bread, white flour and many breakfast cereals, the bran is removed.

The graph below compares the amount of fibre in different types of bread and pasta.



Fibre in bread and pasta

Essential water

Water is another substance that, although not a nutrient, is essential. Water makes up about 60 per cent of your body. The water you lose when you breathe out, sweat and urinate must be replaced. Water is needed to keep your cells working properly and to carry chemicals (including nutrients from food) around your body. Your blood is about 90 per cent water.

A lot of water is taken in by drinking. However, most solid foods contain water. You probably get about one-third of your water from solid food. Raw meat, fruit and vegetables are comprised mainly of water. For example, raw, lean meat and boiled potatoes are both made up of 75 per cent water. If lost water is not replaced, you can become **dehydrated**. A dry throat and mouth is a sign of mild dehydration, which can usually be remedied by drinking plenty of water. Severe dehydration will cause low blood pressure, shock and even death. Dehydration is a major cause of death of children in developing countries, where it is often brought on by diarrhoea.

Try this

ARE ALL BREAKFAST CEREALS THE SAME?

Use the information on the packaging of a range of breakfast cereals to complete the table below. By working in groups, or as a whole class, you should be able to compile quite a large list.

Comparing breakfast cereals

Brand of cereal	Amount per serve (not including milk)					
	Energy (kJ)	Protein (g)	Fat (g)	Total carbohydrate (g)	Sugar (g)	Dietary fibre (g)
e.g. Kellogg's 'Just Right'	478	2.7	0.6	24.2	7.6	3.1

1. Which features of breakfast cereals change the most from brand to brand?
2. Rank the cereals in order from those that contain the most sugar to those that contain the least sugar. Is the order in which the cereals are ranked what you expected?
3. Suggest why some breakfast cereals contain so much more sugar than others.
4. Which brand of breakfast cereal contains the most fibre?
5. The recommended daily intake of fibre is 30–40 g. Can you get all the fibre you need from a breakfast cereal?
6. Why do you think many people still prefer cereals that are low in fibre?
7. Draw a bar graph that compares the amount of fat in the breakfast cereals examined.

Activities

Remember

1. Give two good reasons why you need to eat. Can you think of any other reasons that are not mentioned above?
2. List the five main groups of nutrients.
3. What is fibre?
4. Why is it important to eat fibre even though the chemicals in it are not used by your body?
5. Make a list of foods that you could eat more of to increase your fibre intake.
6. Cereal products like All Bran, Sultana Bran and Bran Flakes are high in fibre. What is bran?
7. What is dehydration and what causes it?

Think

1. Why do most breakfast cereals contain little fibre?
2. Some high-fibre breakfast cereals have more sugar added to them than some of the more highly processed breakfast cereals. Why do you think this is so?
3. Wholegrain products are high in fibre. Why do you think the word 'wholegrain' is used?

Create

Prepare an advertisement to promote increased fibre in people's diets. Your aim is to make high-fibre foods attractive to consumers. Your advertisement could be in the form of a poster, a dramatic performance, on videotape for a television commercial, or on audio tape for a radio commercial.

6.6

Chew it up

Digestion is the breaking down of food so that the nutrients it contains can be absorbed into your blood and carried to each cell in your body.

Four processes are important in supplying nutrients to your cells:

- ingestion — the taking of food into your body
- mechanical digestion
- chemical digestion
- the absorption of the broken-down food into your cells.

Mechanical and chemical digestion

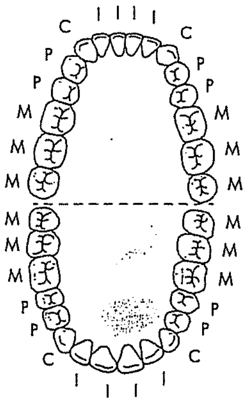
Mechanical digestion involves physically breaking down the food into smaller pieces. Most of this process takes place in your mouth when your teeth bite, tear, crush and grind food. **Chemical digestion** occurs when chemicals in your body react with the food. The chemical reactions that take place change the substances in food into simpler chemicals that are more easily absorbed into your blood.

Chemical digestion is usually assisted by chemicals called **enzymes** that increase the rate of the chemical reactions. Without enzymes, a single meal could take many years to break down. Mechanical digestion increases the rate of chemical digestion because it increases the surface area of the food particles. This exposes more of the food surface to the digestive chemicals and enzymes.

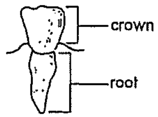
Chemical digestion begins in your mouth where enzymes in saliva begin to break down some of the carbohydrates in the food that you eat.

Look at those teeth!

In many vertebrates, mechanical digestion begins with the teeth. There are four main types of teeth in humans, each type having a different function and position in your mouth. Your teeth are your very own set of cutlery.



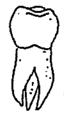
The positions of the different types of teeth



incisors (I) — acting like scissors
- used for biting and cutting food
- chisel edge and broad, flat crown
- found at the front of your mouth

canines (C) — fang-like
- tear and grasp the food
- used for ripping and tearing meat
- sharp point and cone-shaped crown
- found each side of the incisors

molars (M) and **premolars (P)**
- have points or cusps which fit together with those in the upper and lower jaws
- molars generally have between three and five cusps
- premolars generally have two cusps
- used for grinding and crushing food



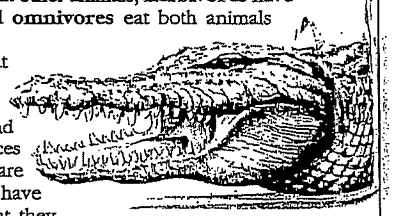
- large surface with hills and valleys
- brick-shaped crown
- found at the back of your mouth

Every tooth tells a tale

The teeth of an animal provide clues about its diet. By examining fossil teeth it is possible to find out about the diets of animals that no longer exist.

Animals can be divided into groups on the basis of what they eat. **Carnivores** are those that eat other animals, **herbivores** have a diet consisting of plants, and **omnivores** eat both animals and plants.

Because the animal flesh that carnivores eat is usually alive and mobile, they need teeth that can hold and kill their prey and then cut it into small enough pieces to swallow. Cats and dogs are carnivores. Most carnivores have well-developed canine teeth that they use to hold their prey, smaller incisor teeth to cut the meat, and molars with special cutting edges that act like scissor blades.



A crocodile's sharp conical teeth are well suited to seizing its prey but not much use for cutting it up. That is why a crocodile spins over in the water with its victim until a part is torn off.

Insectivores are animals that feed on insects. Their teeth are well suited to crushing the tough exoskeletons (external skeletons) of insects. Herbivores, on the other hand, do not need to chase and 'kill' their food because it is stationary. They need teeth that will help them collect food (incisors) and grind it up (premolars and molars). Canine teeth are absent in herbivores. Due to the tough composition of plant food, many herbivores need to spend most of their time grinding their food with their molars. This time-consuming activity wears down their molars. Some herbivores have several sets of molars that act as replacements.

SCI FACTS

It has been reported that a horde of African army ants ate three dead goats in three days. The ants, however, don't always wait for their dinner to die. Tethered horses have been eaten alive so that all that remains is a skeleton. Army ants feed by cutting up their dinner on the spot and then carrying the pieces back home. They may even attack while you sleep. There is a widespread belief in Africa that some types of ants do not attack until they have completely swarmed all over you. They all bite at once when a signal is given by the 'commanding' ant.

Activities

Using data

Copy and complete this table to indicate the numbers of each type of tooth in your own mouth.

	Upper jaw	Lower jaw
incisors		
canines		
premolars		
molars		
Total		

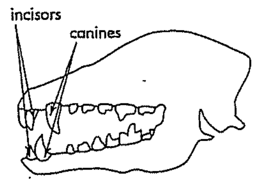
1. What is the total number of teeth in your mouth?
2. Which type of tooth is most common?
3. Draw a map of your teeth like the diagram in the middle of page 136.
4. Compare your map with this diagram. Which teeth are not present in your mouth at the present time?
5. Compare the map of your teeth with that of at least five other students. Record the differences. Suggest some reasons for the differences.

Remember

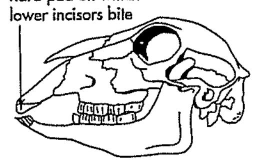
1. How does mechanical digestion assist in chemical digestion?
2. Describe the four important processes involved in getting nutrients to the cells of your body.
3. Explain the difference between mechanical and chemical digestion.
4. What are enzymes and what do they do?
5. Name and describe the four main types of teeth.
6. Describe the function of each of the four types of teeth.
7. Describe the diets of each of the following:
   (a) carnivores
   (b) herbivores
   (c) omnivores.
8. Describe the differences in the teeth of carnivores and herbivores.
9. What is the important job of the teeth of insectivores?

Think

1. Which teeth do you use to bite into a pear?
2. Which teeth are used to crush and grind nuts?
3. Why don't herbivores have canine teeth?
4. The diagrams below show the teeth of a Tasmanian devil and a sheep.



Skull of a Tasmanian devil



Skull of a sheep

(a) Describe the differences between the teeth of these two animals.
(b) How would you expect the diet of a Tasmanian devil to differ from that of a sheep? How did you arrive at your opinion?
(c) What is the likely purpose of the incisors in a sheep?
(d) What are the canines of a Tasmanian devil likely to be used for?

5. How do we know what different types of dinosaurs ate, even though they haven't existed for about 65 million years?

Create

Find out about the feeding habits of an insect of your choice (for example, ant, housefly, bedbug, mosquito or butterfly). Make a model of its feeding parts.

2

25

Teeth

Structure of a tooth

Teeth grow in sockets in the jaw bones. Each tooth consists of a **root**, which is attached to the jaw bone by **cement** and tough fibres; a **neck**, which is the part surrounded by gum; and a **crown**, the part above gum level (Fig. 1). The crown is the biting surface. It has an outer layer of **enamel** which is the hardest substance in the body. The remainder of a tooth consists of a bone-like material called **dentine**, except for the central **pulp cavity** which contains nerves and blood vessels. Figure 2 illustrates the arrangement and types of teeth.

Milk and permanent teeth

Teeth develop inside the jaws before birth and begin to appear, or erupt, at about five months of age. By about age six years most children have twenty-four teeth. Twenty of these are **milk teeth**. These are shed between the ages of seven and eleven years and are replaced by larger **permanent teeth** (Fig. 3). By eighteen years of age most people have a total of thirty-two teeth. The last to appear are the rear molars, or wisdom teeth.

Care of teeth

Decay occurs because bacteria in the mouth change carbohydrate food stuck to teeth into acids which dissolve away first the enamel and then the dentine (Fig. 4A). Tooth decay can be avoided by brushing at the very least once a day in a way which removes all the food, especially that between the teeth. A side-to-side and up-and-down scrubbing action with the brush is *not* recommended because it leaves food between the teeth and can damage gums. Figure 4B illustrates one of the commonly recommended methods for thoroughly cleaning teeth.

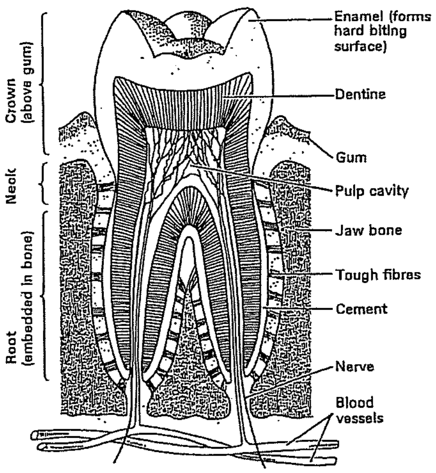


Fig. 1. Structure of a tooth. The root of a tooth is attached to the jaw bone by cement and tough fibres. The crown is the biting surface of a tooth, and is made of enamel. The inner part of a tooth is made of bone-like material called dentine, and contains a pulp cavity filled with nerves and blood vessels.

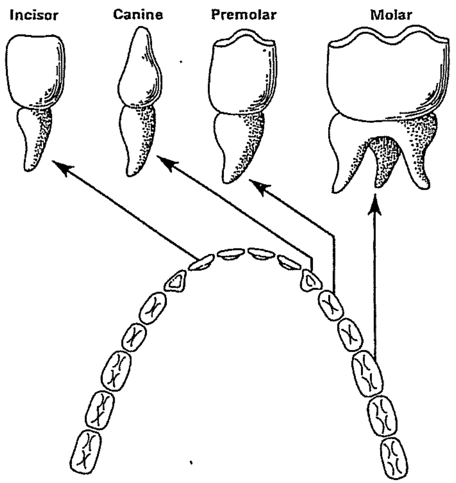
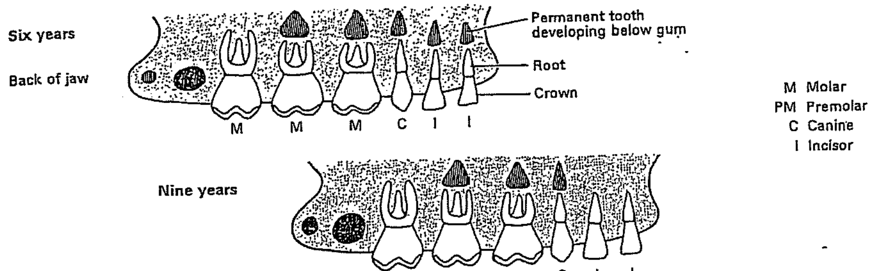


Fig. 2. There are four types of tooth. Incisors and canines are used for biting. Premolars and molars are used for chewing, which crushes and pulverizes food making it ready for digestion.



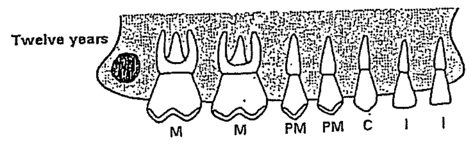


Fig. 3. These drawings show teeth in half of the upper jaw. Teeth develop inside the jaws before birth, and appear about five months after birth. Milk teeth are later replaced by permanent teeth. Find milk teeth in the drawings by looking for those which have a permanent tooth developing above them.

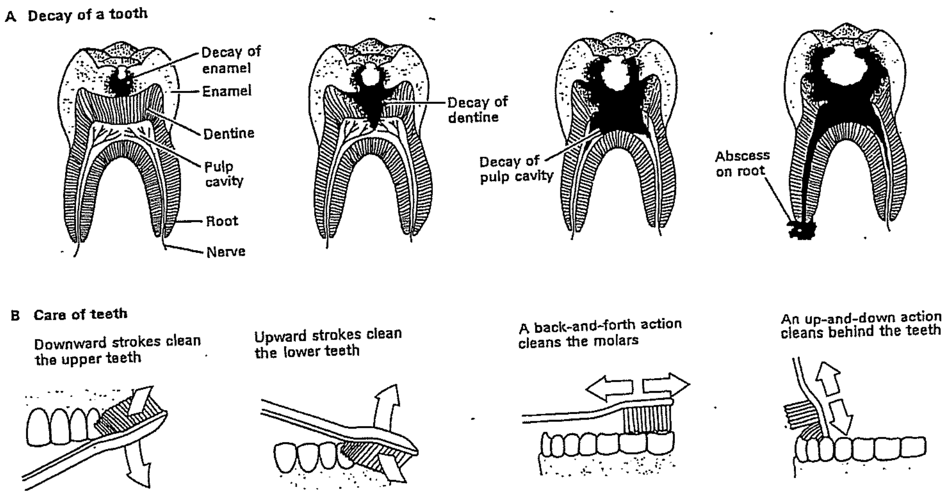


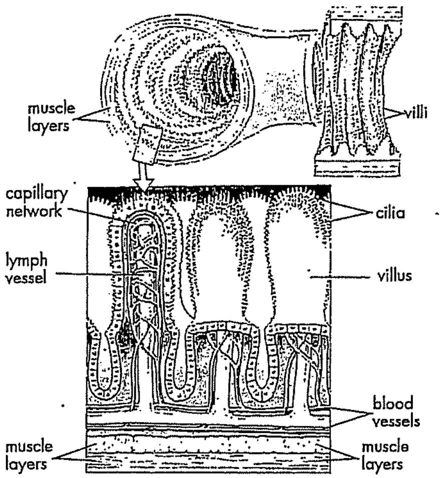
Fig. 4. Care of teeth. Bacteria in the mouth digest food stuck to the teeth, changing it to acid which dissolves enamel and dentine. Brush the teeth as shown above at least once a day to remove all the food and bacteria on and inbetween the teeth.

3

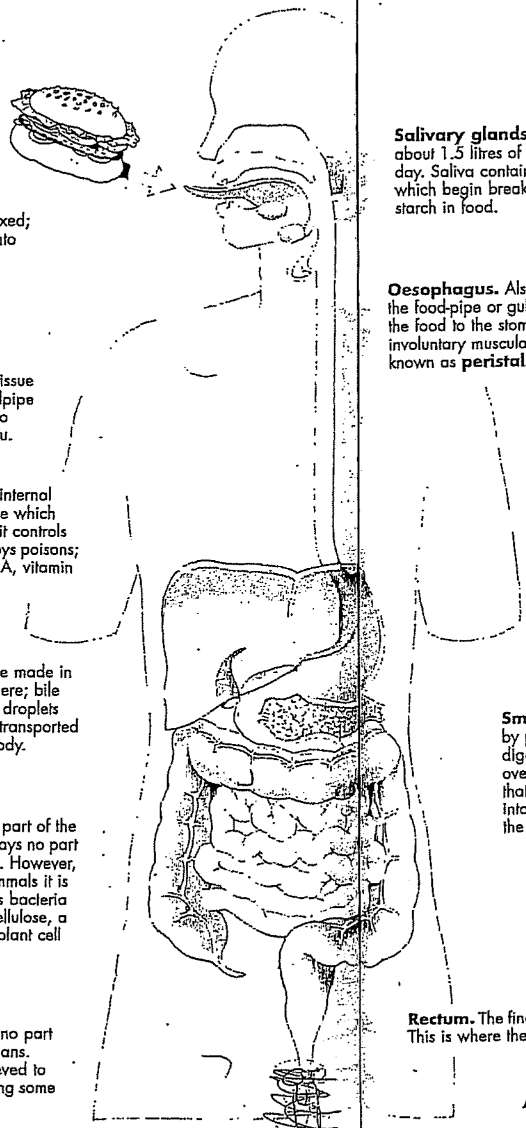
6.7 Down we go . . .

digestive

Your main digestive highway is called the **alimentary canal**. It consists of a long tube with coils, large caverns and thin passageways. Other organs that provide chemicals to break down the food or absorb nutrients are attached to the alimentary canal. The alimentary canal begins at the mouth and ends at the anus.



The absorption of most nutrients into your body occurs in the ileum, the last section of the small intestine. The finger-like villi on its walls give it a large surface area that speeds up nutrient absorption. Many tiny blood vessels called **capillaries** transport the nutrients from the villi into your bloodstream. Undigested material continues on to the large intestine where water and vitamins may be removed, and then the remainder is pushed out through the anus as faeces.



Mouth. Food and saliva are mixed; teeth mechanically break food into smaller pieces.

Epiglottis. A flap of tissue that closes off the windpipe so that food doesn't go down it and choke you.

Liver. The largest internal organ. It makes bile which breaks down fats; it controls blood sugar; destroys poisons; and stores vitamin A, vitamin D and iron.

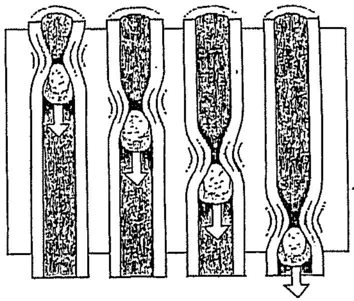
Gall bladder. Bile made in the liver is stored here; bile breaks up fats into droplets small enough to be transported to the rest of the body.

Caecum. The first part of the large intestine, it plays no part in human digestion. However, in plant-eating mammals it is larger and contains bacteria that break down cellulose, a chemical found in plant cell walls.

Appendix. Plays no part in digestion in humans. However, it is believed to play a role in fighting some diseases.

Salivary glands. Make about 1.5 litres of saliva each day. Saliva contains enzymes which begin breaking down starch in food.

Oesophagus. Also known as the food-pipe or gullet. It carries the food to the stomach, using involuntary muscular contractions known as **peristalsis**.



Partly digested food is forced along the oesophagus by peristalsis — a wave of involuntary muscular contractions.

Stomach. A temporary food storage area, which can expand to hold between 2 and 4 litres of food. Muscle movements in the stomach wall mix the food with gastric juice, which helps to break down proteins. The stomach also contains dilute hydrochloric acid which kills germs and provides a suitable environment for protein digestion.

Pancreas. Makes pancreatic juice which is alkaline (base dissolved in water), so it neutralises the stomach acid. Enzymes that break down proteins, fats and carbohydrates are also made here.

Small intestine. A tube about 6 m long. Food moves through it by peristalsis. The small intestine makes enzymes that complete the digestion process. The cells in the wall of the small intestine release over 5 litres of mucus and water each day. It is in the small intestine that nutrients from now almost totally digested food are absorbed into your bloodstream. The blood then carries the nutrients to all of the cells of your body.

Large intestine (colon). Undigested material passes into the large intestine. As the material is pushed through it by peristalsis, water, salts, vitamins and any remaining sugars are absorbed so that they can be reused by the body.

Rectum. The final part of the large intestine. This is where the faeces are stored.

Anus. The faeces pass through here when you go to the toilet.

Activities

Remember

1. List each of the major organs that are part of the alimentary canal, or are attached to the alimentary canal, starting at the mouth. State the role that each of these organs takes in the process of digestion.
2. What is peristalsis and why does it occur?
3. Where does digested food eventually end up?
4. Where does undigested food eventually end up?
5. Where are villi found and what do they do?

Think

1. Why is it necessary for the body to break down the food that we eat?
2. Why is the caecum larger in herbivores?
3. Some of the organs used in digestion are part of the alimentary canal, while others are attached to the alimentary canal. List those that are actually part of the alimentary canal.

Create

1. Write a script for a play in which a group of people can show how the digestive process occurs and what happens to food in your alimentary canal.
2. Build a 'working' model of a digestive system.
3. Design a board game that shows how the human digestive system works.

Research

1. Find out where the anal sphincter is and what it does.
2. Research and report on the digestive system of an animal of your choice, and then present your findings to the class. Choose from the following animals: rabbit, cow, dog, earthworm or snake.

24

Digestion, absorption, and assimilation

Digestion

Digestion is the process which makes food soluble. All foods must be made soluble before they can pass into the bloodstream and be transported around the body to all the cells.

Food is digested inside a tube called the **alimentary canal**. This tube starts at the mouth and ends at the anus. First the food is broken down into small pieces by the teeth. This is called mechanical breakdown of food. Next, chemical breakdown of food occurs as follows. In certain places the walls of the alimentary canal produce digestive juices containing chemicals called **digestive enzymes**. These enzymes break food down into chemically simpler substances which are soluble. One theory of how this happens is explained in Unit 4.

Digestive enzymes can be divided into groups according to the type of food they digest. **Amylases** are a group of enzymes which digest starchy foods into sugars such as glucose. **Lipases** are enzymes which digest fats and oils into simpler substances called **fatty acids** and **glycerol**. **Proteases** digest proteins into simpler substances called **amino acids** (Fig. 1). When digestion is complete absorption takes place.

Absorption

Absorption is the movement of digested (soluble) foods through the walls of the alimentary canal into the bloodstream. The food is transported first to the liver and then to all the other cells of the body, where it is assimilated.

Assimilation

Assimilation is the name for the processes by which cells take in and make use of digested food to produce energy, and as a raw material for growth and repair of tissues—in other words, for metabolism.

Most foods contain substances which cannot be digested. Humans, for instance, have no enzymes which can digest the cellulose cell walls in foods taken from plants. Cellulose and other indigestible materials are called **faecal matter**, or **faeces**. Faeces pass out of the body through the anus. This process is called **defecation** (Fig. 2).

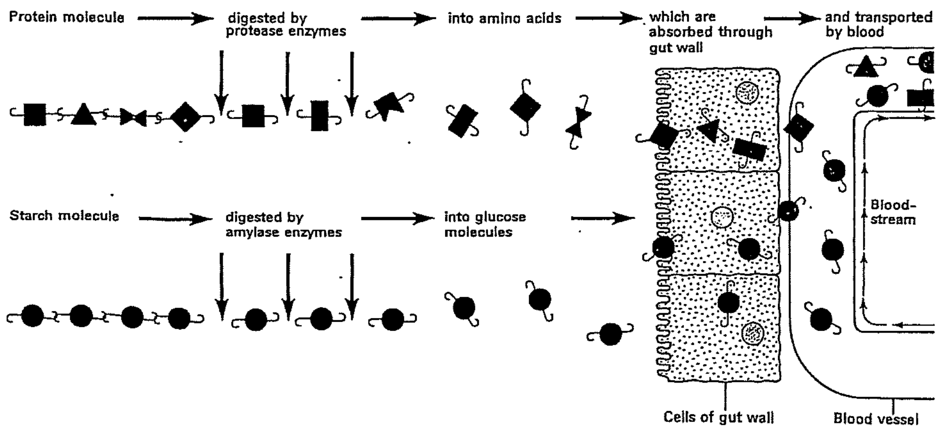


Fig. 1. Digestive enzymes digest food by breaking it down into chemicals which are soluble. Food then passes through the gut wall into the bloodstream—a process called absorption.

Blood transports food to all parts of the body where it is used (assimilated) by cells.

5

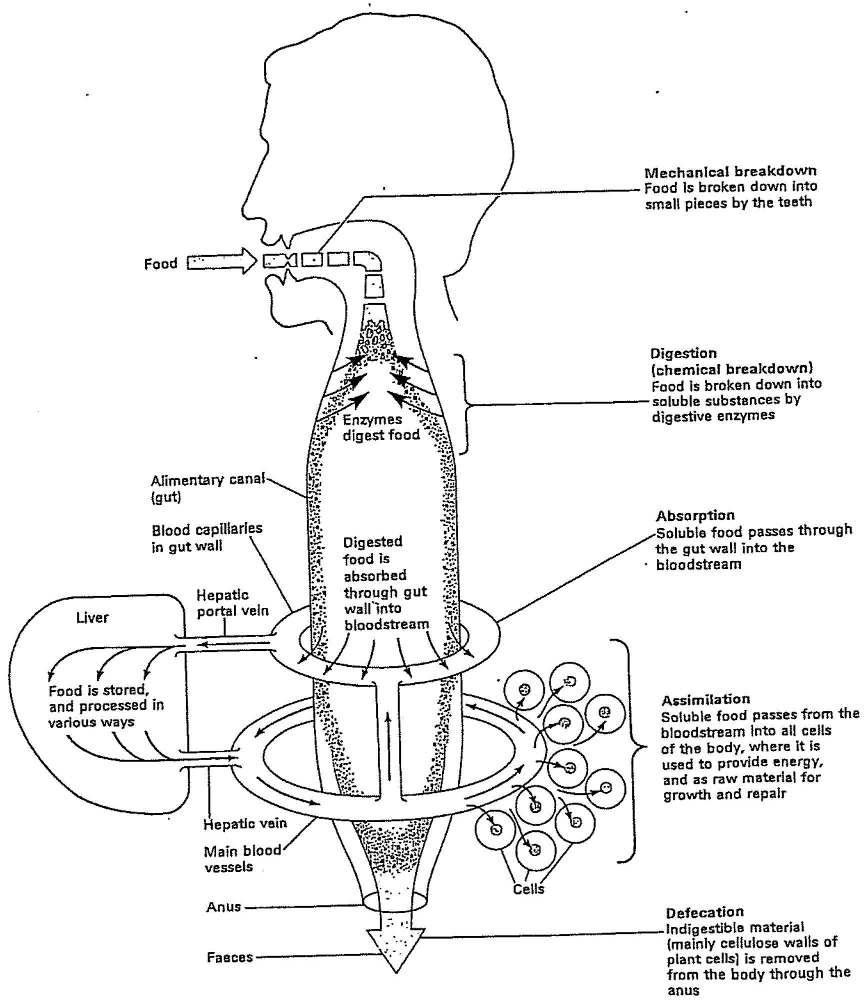


Fig. 2. Food is digested inside a tube called the alimentary canal, or gut—shown here as a straight tube from the mouth to the anus. First, food is broken into small pieces by the teeth. Second, enzymes break down food into soluble chemicals. Third, the soluble food is absorbed through the gut wall into the bloodstream. Fourth, food is transported to the liver and then to all parts of the body in the blood. Fifth, the food is assimilated—it is used in the cells to provide energy, and as material for growth and repair.

26

Digestion

Digestion in the mouth

Chewing mixes food with **saliva**, produced by the salivary glands. Saliva moistens food so that it can be swallowed easily and it contains an enzyme, **salivary amylase**, which breaks down cooked starch into a sugar called maltose. During swallowing food is formed into a ball, or **bolus**, by the tongue and pushed to the back of the mouth. The soft palate is pushed upwards preventing food from entering the nasal cavity and the entrance to the wind-pipe is also closed (Fig. 1). The bolus then passes down the throat into the **oesophagus**, or gullet. Muscular contractions called **peristalsis**, described in Fig. 2, force food down into the stomach, and throughout the remainder of the digestive system.

Digestion in the stomach

The stomach has rings of muscle at both ends called **sphincters** (Fig. 4). After a meal both sphincters contract and keep food in the stomach for about an hour. Muscles in the stomach wall contract rhythmically, mixing food with a liquid called **gastric juice**. This contains **pepsin**, an enzyme which digests protein, and a weak solution of hydrochloric acid, which kills germs and helps the enzyme to work. When the lower sphincter opens, partly digested food moves into the **duodenum**, which is the first part of the **small intestine**.

Digestion in the small intestine

In the duodenum food is mixed with **bile**. Bile s made in the liver, which is described in Unit 28. 3ile helps digestion by changing fats and oils into an :mulsion, which means they are broken down into ninute droplets. Bile also neutralizes stomach acid. This enables enzymes in the small intestine to work. 'ood is mixed with enzymes from the pancreas, and iter with more enzymes from the ileum wall (Fig.). These enzymes digest food completely, which is ien absorbed into the bloodstream.

Functions of the large intestine

The substances in food which cannot be digesd pass into the large intestine. Here, water and salt e removed. The remaining faecal matter passes out the body through the anus.

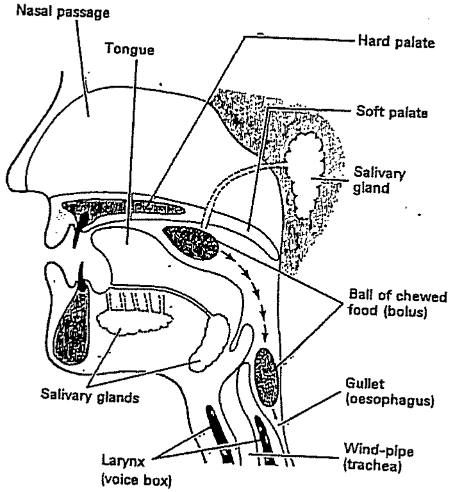


Fig. 1. Digestion in the mouth and swallowing. Chewing mixes food with saliva from the salivary glands. Saliva moistens food and contains an enzyme which breaks down cooked starch into maltose sugar. During swallowing the tongue pushes food to the back of the mouth from where it passes into the gullet. The soft palate prevents food from entering the nasal cavity and the entrance to the wind-pipe is also closed.

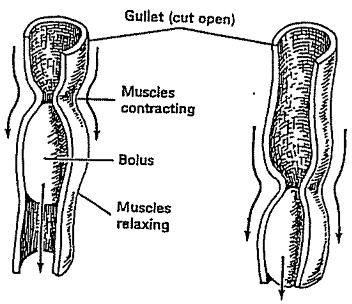


Fig. 2. Waves of muscular contraction called peristalsis move food along the gut. Circular muscles in the gut wall contract behind the food and relax in front of it, pushing it forwards at about 20 cm a second.

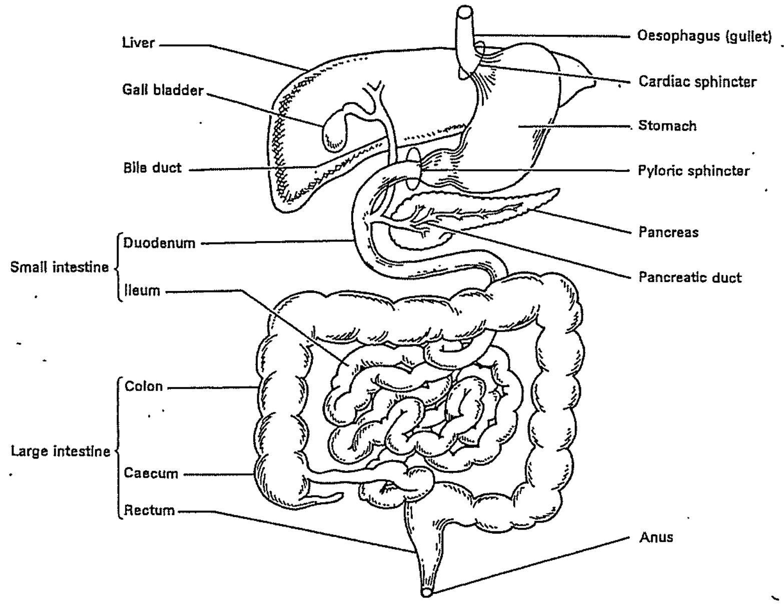


Fig. 3. The alimentary canal (gut) partly unravelled to show the pancreas.

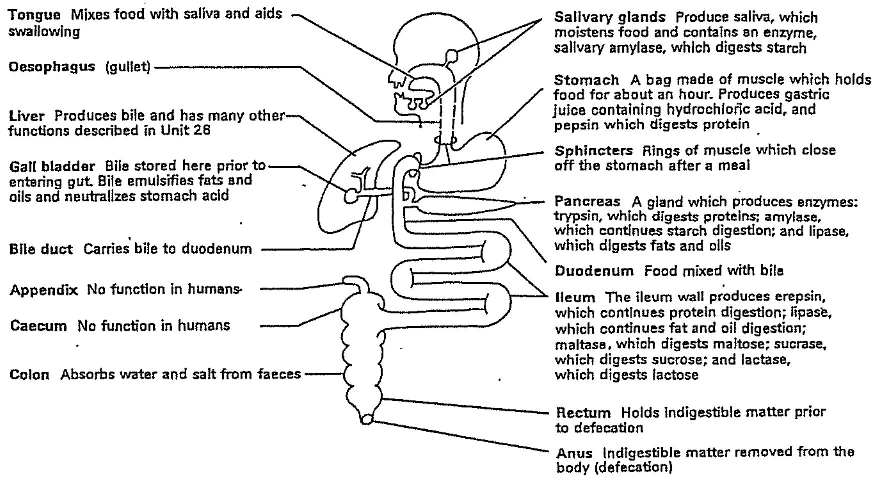


Fig. 4. Diagram of the gut and functions of its parts.

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Absorption

Digestive enzymes break down food into a liquid called chyle. This liquid is a mixture of many substances. It contains glucose and other sugars produced by digestion of carbohydrates; amino acids produced by protein digestion; fatty acids and glycerol from fat and oil digestion; and some undigested but emulsified fats. Chyle also contains vitamins and minerals. Nearly all absorption of this digested food takes place in the ileum of the small intestine.

The whole of the ileum's internal surface is covered by finger-like projections called **villi** (*singular*: villus) which are about 1 mm long (Figs. 1 and 2). Each square millimetre of ileum wall has up to forty villi and there are about five million in the ileum as a whole. The presence of villi gives the ileum a far greater internal surface area available for absorption than if it had a smooth lining. Originally it was thought that this area was about ten square metres in humans, but under very high magnification (×40 000) the surface of each individual cell lining the ileum is seen to be folded into **microvilli** (Fig. 2B), giving the ileum an estimated total internal surface area of thirty square metres.

Inside each villus there is a dense network of blood capillaries, and a single **lacteal**, or lymph vessel, which is closed at its upper end (Figs. 1 and 2B). After digested food has been absorbed into the villus, it passes into these two kinds of vessels.

Amino acids, sugars, vitamins, minerals, and small amounts of fatty acids and glycerol pass into the blood capillaries of the villi. These capillaries join together to form a large blood vessel called the hepatic portal vein which carries the food to the liver. What happens to this food next is described in Unit 28. The bulk of the fatty acids and glycerol pass through the villi walls into the lacteals. The remaining emulsion of oil droplets is also absorbed into the lacteals of the villi. This happens in the following manner. The oil droplets pass between the microvilli of the cells covering each villus and are absorbed whole into these cells. The droplets then pass through the cytoplasm of the cells and out of the other side into the lacteals. These oil droplets, together with more droplets built up from absorbed fatty acids and glycerol, pass into the main lymphatic system and are eventually discharged into the bloodstream, as described in Unit 34.

Movement of absorbed food out of the villi into the bloodstream and lymphatic system is assisted by periodic contractions of villi, during which they suddenly become shorter and fatter and then relax slowly. This happens about six times a minute in each villus.

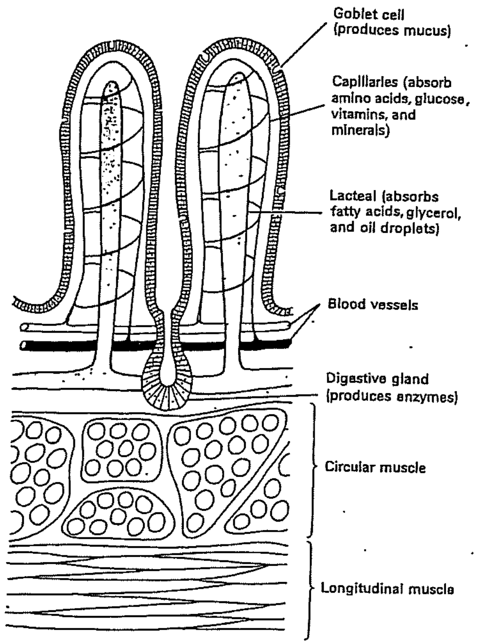


Fig. 1. Diagram of the ileum wall. Soluble (digested) food is absorbed in the ileum, through the surface of finger-like projections called villi. The presence of villi gives the ileum a far greater internal surface area for absorption than if it had a smooth lining. Amino acids, sugars, vitamins, and minerals are absorbed into blood capillaries inside each villus. The blood carries these foods to the liver and then distributes them around the body. Fatty acids, glycerol, and tiny oil droplets are absorbed into a lacteal at the centre of each villus. Then they are transported through the lymphatic system and into the bloodstream.

7

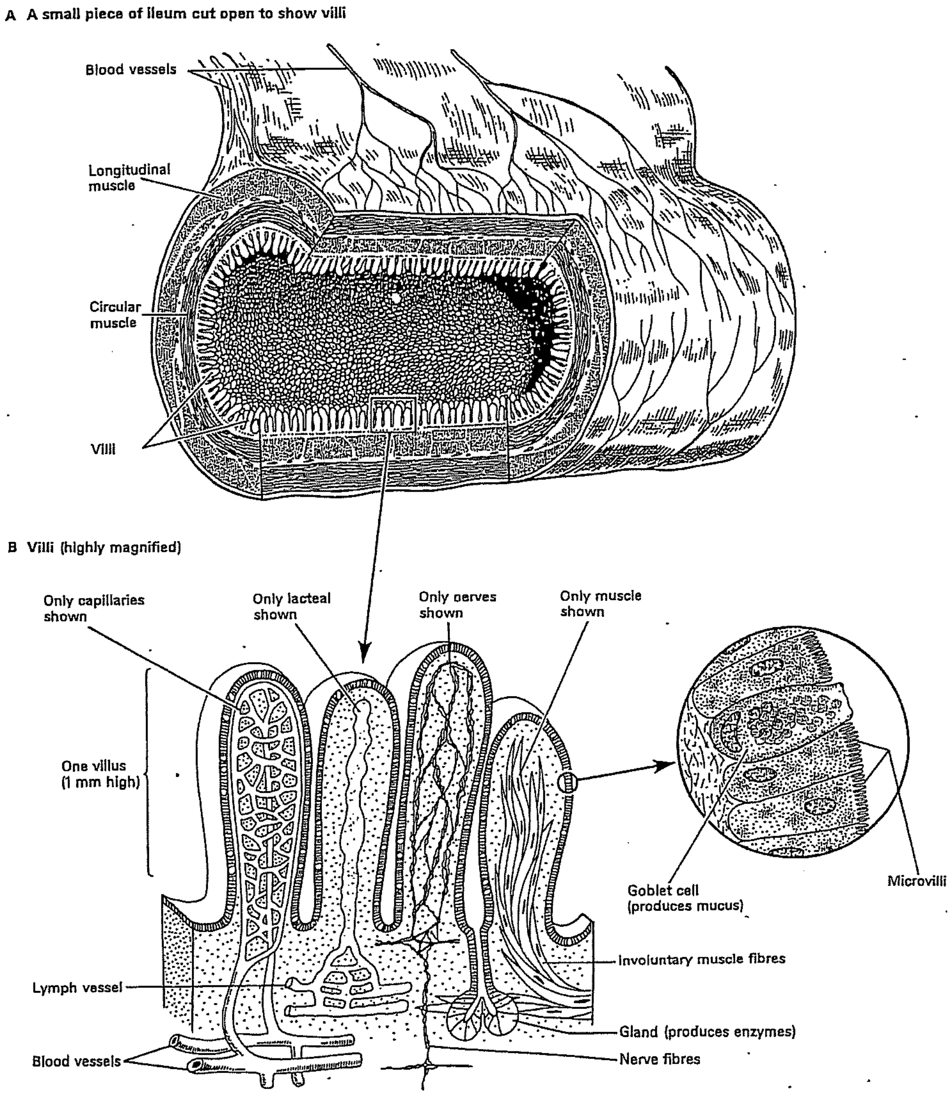


Fig. 2. Structure of the ileum wall.

6.3 Keeping healthy

Although carbohydrates, fats and oils provide you with energy, they alone will not keep you healthy. Proteins, vitamins and minerals provide you with chemicals needed for growth, repair and protection against disease. There are government regulations that require manufacturers to provide basic information about the nutrients in packaged foods. This information makes it easier for you to select healthy food.

Practical TESTING FOR PROTEIN

AIM
To identify protein with a chemical test and to test for it in foods

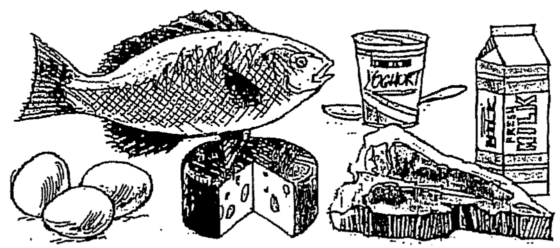
YOU WILL NEED
gelatine solution
copper sulfate solution
dilute sodium hydroxide
food samples
mortar and pestle
6 test tubes and test tube rack

- Place a few drops of gelatine solution in a clean test tube.
- Place a few drops of water into another clean test tube.
- Place two drops of copper sulfate solution into each of the two test tubes. Then add two drops of sodium hydroxide solution to each.
- Gently swirl each test tube so that the liquids mix. Look for any colour changes that take place. Record your observations.

Gelatine is a source of protein. The test that you have conducted is called the Biuret test and can be used to find out whether protein is present in food.

- Use the Biuret test to find out whether protein is present in a selection of foods. If the food is solid, carefully use a mortar and pestle to mash a pea-sized amount in some water. Make sure that each test tube is clean before you conduct the test. You should test foods such as milk, meat, fruits, vegetables, bread and snack foods.

1. Which foods tested appear to contain protein?
2. According to the illustration below, was the Biuret test successful?



The main food sources of protein

Proteins are chemicals that are essential for the growth and repair of cells. All living matter contains proteins. Proteins are made up of simpler building blocks called **amino acids**, which contain the elements carbon, hydrogen, oxygen and nitrogen. Extra protein is needed by pregnant women, breastfeeding mothers, growing children and adolescents.

and

Vitamins are substances that you need in small amounts. They are needed to control some of the chemical changes that occur in the cells of your body. Some vitamins are actually made by your body. Most, however, are found in the food you eat. A lack of any of the thirteen vitamins is likely to result in disease. Diseases caused by a lack of vitamins are called **vitamin deficiency diseases**, and include scurvy, beriberi and rickets. These diseases have become less common as people have become more aware of the importance of vitamins.

Minerals are chemical elements that are needed in small amounts. Like vitamins, they help to control some of the chemical changes that occur in your body. A lack of important minerals is likely to restrict healthy growth and cause disease.

Vitamins and minerals are often added to foods to make them more healthy or attractive to the consumer. Some minerals that used to be added as preservatives or to add flavour (such as salt) are no longer added.

Some vitamins needed for good health

Vitamin	Why it is needed	Sources
A	healthy skin and eyes; lining of digestive and respiratory systems; healthy bones and teeth	fish liver oils, butter, margarine, green and yellow vegetables, carrots, tomatoes, liver, kidneys, milk, eggs
B1 (thiamine)	the release of energy from carbohydrates; helps the heart work properly	seafood, meat, wholegrain breads and cereals, nuts, peas, potatoes, most vegetables
B2 (riboflavin)	healthy skin and eyes; repair of damaged tissue; helps cells use oxygen	meat, fish, eggs, green vegetables, mushrooms, pasta, wholegrain breads and cereals, milk, cheese
B3 (niacin)	healthy skin; helps release of energy from carbohydrates; fats and proteins	leafy vegetables, wholegrain breads and cereals, peanut butter, potatoes, tuna, eggs, liver
B6	healthy teeth and gums; red blood cells and blood vessels	yeast, wholegrain bread and cereals, meat, soya beans, egg yolk, peanuts, most vegetables
B12	proper development of red and white blood cells	green vegetables, liver, meat, eggs, fish, milk and milk products
C (ascorbic acid)	healthy bones and teeth; healthy tissues; healing of wounds	citrus and other fruits, tomatoes, leafy vegetables, potatoes
D	allows calcium and phosphorus to be used, to help bones and teeth grow and stay healthy	liver, fish, eggs; also made in the skin when it is in sunlight
E	keeps body cells healthy	butter and margarine, bread, green leafy vegetables, wholegrain cereals
K	normal blood clotting	green vegetables, tomatoes, cabbage, cauliflower, potatoes, cereals, eggs; also made by bacteria in the large intestine

Some minerals needed for good health

Mineral	Why it is needed	Sources
calcium	healthy bones and teeth; blood clotting	milk, cheese, yoghurt, green vegetables
fluorine	hardens tooth enamel	drinking water
iron	needed by the red blood cells that carry oxygen around the body	red meat, liver, wholegrain bread and cereals, green vegetables
iodine	to help control body growth	fish, iodised salt
phosphorus	healthy bones and teeth; proper working of nerves and muscles	milk and milk products, wholegrain bread and cereals, most vegetables
sodium	proper working of nerves; keeps blood healthy	table salt (sodium chloride), vegetables
potassium	proper working of nerves; keeps blood healthy	meat, apricots, bananas, potatoes, milk, vegetables
zinc	healing of wounds; proper growth; vision	liver, onions, green vegetables

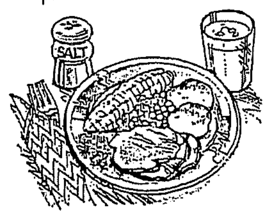
Activities

Remember

1. Why is protein an important nutrient?
2. What types of food would you recommend to a person lacking:
   (a) vitamin C
   (b) calcium
   (c) iron?

Think

1. Examine the meal pictured on the right.
   (a) Use the information in the tables to list the vitamins and minerals that are not present in this meal.
   (b) What are the sources of protein in this meal?



2. Why do pregnant women, children and adolescents need more protein?
3. Milk and other dairy products are well known as good sources of calcium. Which nutrients would be missing from the diet of someone whose food intake consisted mainly of dairy products?
4. Too much salt (sodium chloride) in your diet is not healthy. Why do we need salt at all?

Create

Produce a colourful poster that illustrates the benefits of fruit or fresh green vegetables in your diet.

8

6.4 Keeping a balance

In order to grow and stay healthy, you need a diet that contains enough of each of the essential nutrients. You also need an adequate supply of fibre and water.

Fruit and vegetables are good for you. However, if you ate only fruit and vegetables you would be unlikely to get enough protein to keep yourself healthy. If you ate only meat and eggs you would be getting lots of protein but would be lacking some vitamins, carbohydrates and fibre.

No single food can supply every essential nutrient. You need a balanced diet that will provide you with all the necessary chemicals and enough energy to get you through each day. Food that we describe as 'junk food' generally contains a lot of fats and carbohydrates, but very little protein, vitamins or minerals. Much of this type of food also contains large amounts of unnecessary salt.

SCI FACTS

Many animals have very odd eating habits. Rabbits eat their own droppings, mice eat soap and the vampire moth sucks blood from other animals. The female praying mantis eats its male partner while they are mating. The unfortunate male continues mating with its partner even after its brain has been eaten.

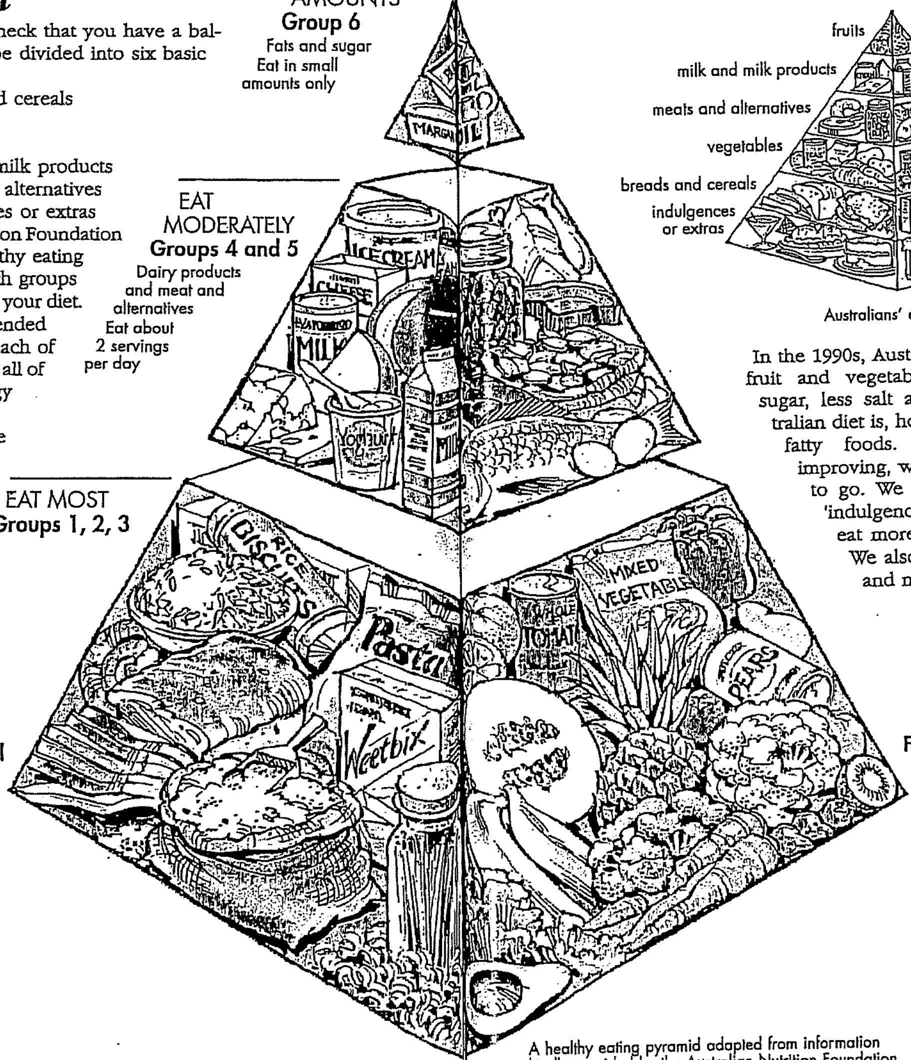
healthy eating pyramid

To make it easier to check that you have a balanced diet, food can be divided into six basic food groups.

- Group 1: Breads and cereals
- Group 2: Vegetables
- Group 3: Fruits
- Group 4: Milk and milk products
- Group 5: Meats and alternatives
- Group 6: Indulgences or extras

The Australian Nutrition Foundation has constructed a 'healthy eating pyramid' to show which groups need more emphasis in your diet. By eating the recommended amount of food from each of the groups, you can get all of the nutrients and energy you need.

The food groups at the top of the pyramid are those that you need a little of each day. The food groups at the bottom of the pyramid are those that you need a lot of each day.



EAT SMALL AMOUNTS
Group 6
Fats and sugar
Eat in small amounts only

EAT MODERATELY
Groups 4 and 5
Dairy products and meat and alternatives
Eat about 2 servings per day

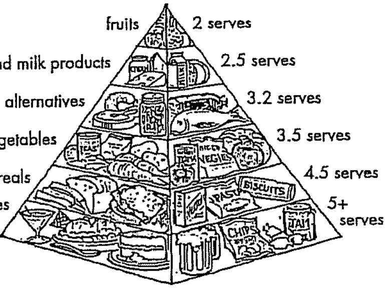
EAT MOST
Groups 1, 2, 3

Bread and cereal products
Eat at least 4 servings per day

Fruits and vegetables
Eat at least 4 servings per day

A healthy eating pyramid adapted from information kindly provided by the Australian Nutrition Foundation

Australians in general do not eat a balanced diet. However, our diets are improving! A survey conducted by the CSIRO in 1993, sponsored by Edgell-Birds Eye, revealed that Australians in the 1990s eat a healthier diet than they did in the 1980s.



Australians' eating habits

In the 1990s, Australians are eating more fruit and vegetables, more fibre, less sugar, less salt and less fat. The Australian diet is, however, still too high in fatty foods. While our diet is improving, we still have a long way to go. We need to cut down on 'indulgences and extras' and to eat more fruits and vegetables. We also need to eat less meat and meat products.

Activities

Remember

1. What is meant by the term 'balanced diet'?
2. Fruit and vegetables are nutritious foods. Why can't you live healthily on fruit and vegetables alone?
3. Draw your own balanced diet pyramid. List or draw pictures of some appropriate foods on each level of the pyramid.

Think

1. The Australian diet is too high in fatty foods. Which foods do you think are the cause of this?
2. Why are the diets of most Australians becoming healthier?
3. Can a vegetarian have a balanced diet? What foods would be needed to make up for the lack of meat?

Investigate

1. Design a table in which you can record everything you eat and drink for three days. You will need a column or row for each of the five food groups. Complete the table progressively over a three-day period.
2. How does your food intake compare with the balanced diet pyramid on each of the three days?
3. Use your table to complete the following checklist:

Am I getting enough protein? (groups 4 and 5)	
Am I getting enough carbohydrates? (groups 1 and 3)	
Am I getting enough fibre? (groups 1, 2 and 3)	
Am I getting enough calcium? (group 4)	
Am I getting all necessary vitamins and minerals? (all groups)	

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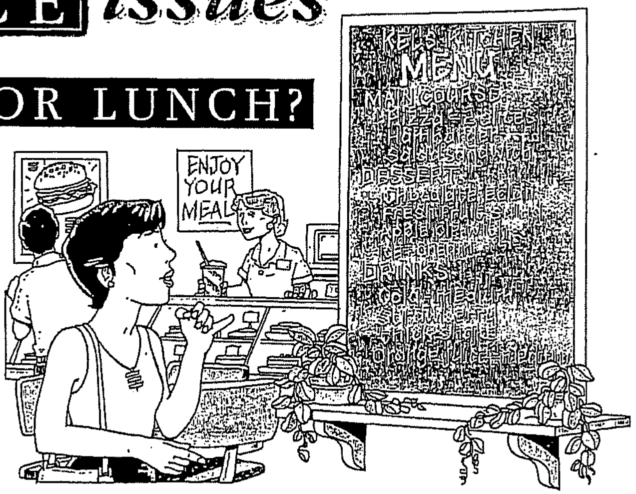
6.5 SCIENCE issues

WHAT'S FOR LUNCH?

Rebel, an active and growing 13-year-old girl, has just played a game of netball and is about to join some friends at a movie. First, however, she wants a quick but nutritious lunch and she decides to have lunch at the local snack bar, Kel's Kitchen. The menu is not exactly inspiring but the food is cheap and service is quick.

Rebel selects a lunch that she will enjoy. The nutritional information for the snack bar menu and for Rebel's lunch is shown in the tables below.



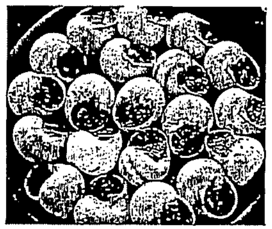
Nutrition information for the snack bar menu

Food	Energy (kJ)	Carbo-hydrates (g)	Fats (g)	Protein (g)	Calcium (mg)	Iron (mg)
pizza (2 slices)	2060	62	20	16	36	2.6
hamburger	1900	37	20	27	56	4.2
salad sandwich	940	31	9	5	13	0.8
chocolate eclair	1320	39	15	8	90	1.3
fresh fruit salad	290	18	0.3	1	17	0.6
apple pie with ice-cream	2310	78	26	6	93	0.6
medium cola	384	25	0	0	0	0
strawberry thick shake	1230	27	15	12	390	0.3
medium orange juice	530	31	trace	2	26	0.2

Rebel's lunch

Food	Energy (kJ)	Carbo-hydrates (g)	Fats (g)	Protein (g)	Calcium (mg)	Iron (mg)
hamburger	1900	37	20	27	56	4.2
apple pie with ice-cream	2310	78	26	6	93	0.6
strawberry thick shake	1230	27	15	12	390	0.3
Total	5440	142	61	45	539	5.1

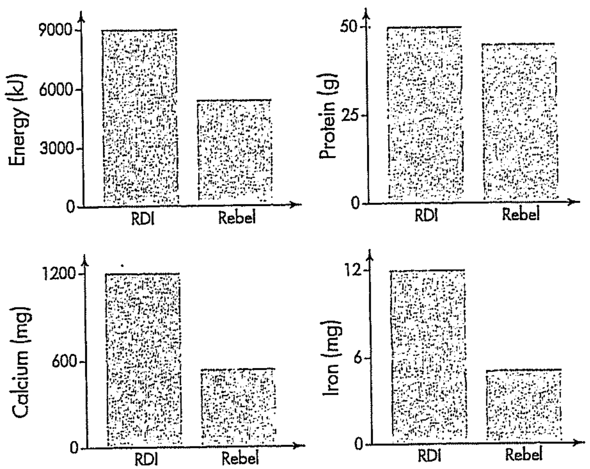
SCIFACTS



Snails filled with garlic butter for sale in France

What's for lunch in other countries? In France you might enjoy some snails with garlic butter. About 40 000 tonnes of snails are eaten in France each year. Or while in France you might find frogs' legs more appetising. But maybe you prefer to feast on bats in Indonesia or locusts in Burma.

Rebel then constructs four graphs to compare her meal with the recommended daily intakes for energy and three of the nutrients.



Comparing Rebel's meal with recommended daily intake (RDI)

Approximate recommended daily nutrient intakes for 12–15 year olds

Nutrient	Recommended daily intake	
	Males	Females
protein (g)	50	50
calcium (mg)	1200	1000
iron (mg)	12	12

Recommended daily energy intake, in kilojoules, for 12–15 year olds

Age (years)	Recommended daily intake (kJ)	
	Males	Females
12	9800	8600
13	10 400	9000
14	11 200	9200
15	11 800	9300

Activities

Remember

What do the letters RDI stand for?

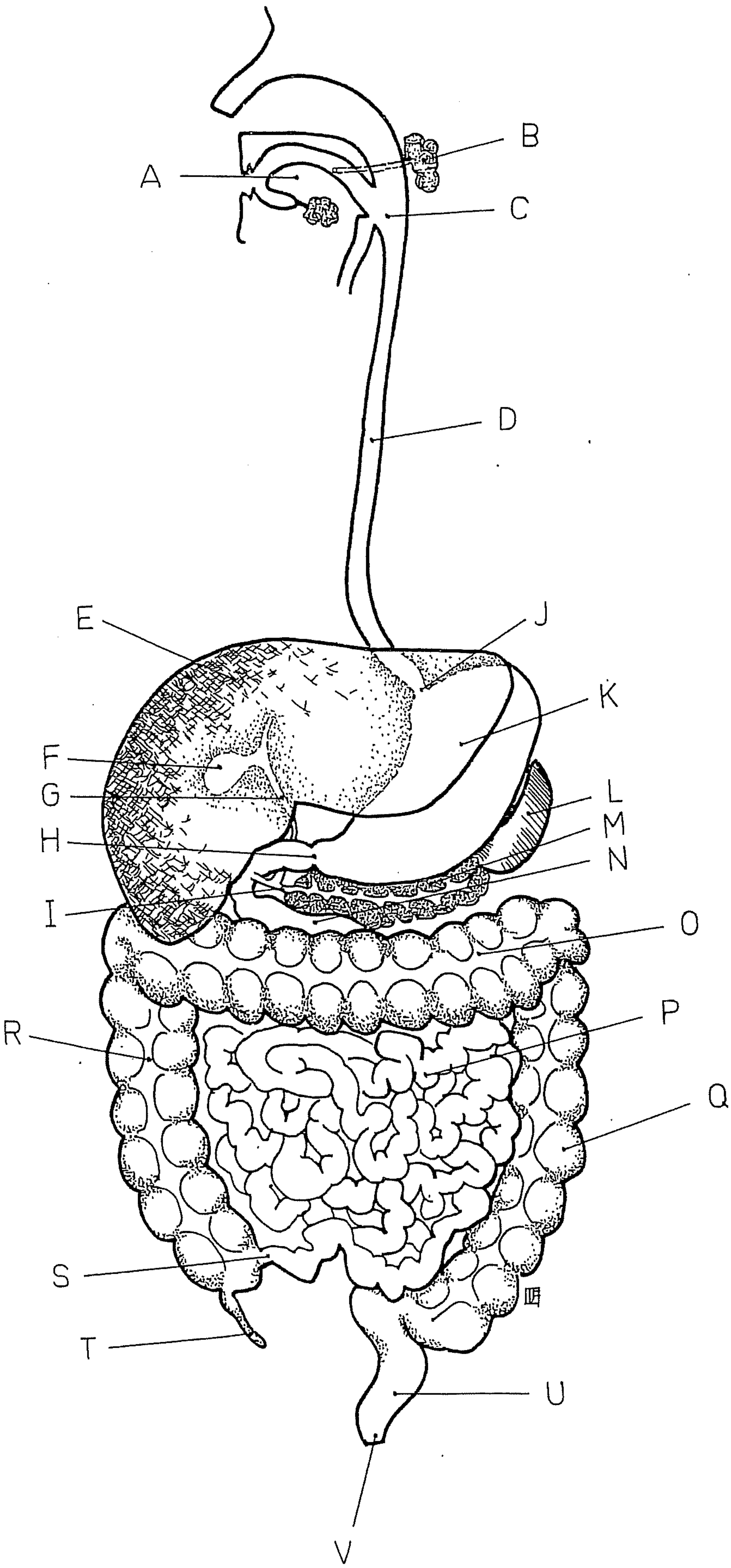
Using data

1. Which nutrient is Rebel getting most of, compared with her recommended daily intake? What is the benefit to Rebel of a high intake of this nutrient?
2. Rebel's recommended daily energy intake is 9000 kilojoules. What advice would you give Rebel about her choice of food for breakfast, dinner and snacks?
3. Select two separate meals from the snack bar menu for yourself.
   (a) Meal 1: Your most appealing lunch
   Select a main course, dessert and drink that you would most enjoy eating.
   (b) Meal 2: Your most nutritious lunch
   Select a main course, dessert and drink that you feel would provide you with the healthiest lunch.
4. Record your selections in tables like Rebel's (see the second table): one headed 'most appealing lunch' and the other 'most nutritious lunch'. Fill in all the energy and nutrients in your tables.
5. Calculate the total amount of energy and nutrients for each meal.
6. Compare your totals with the recommended daily intakes by drawing four bar graphs like Rebel's. Each of your graphs will include three bars: one for your RDI, one for your most appealing lunch and one for your most nutritious lunch. Your RDI can be found in the third and fourth tables.
7. Use your graphs to answer the following questions.
   (a) Which meal would give you the most energy?
   (b) Would your most nutritious meal provide you with enough energy for the day? If not, what could you do about it?
   (c) Which meal would give you the most of each of the three nutrients?
   (d) Which nutrient does your most appealing dinner give you the most of, when compared to your RDI?
   (e) Remembering that you will probably have eaten two other meals on this day, should you still eat your most appealing meal? If you were to eat your most appealing meal, what would you need to be careful about in choosing your other meals?

Research

Some fast-food chains provide pamphlets with nutritional information about their products. Collect some information about fast food and write a report on its nutritional value.

DIGESTIVE SYSTEM



A ....
B ....
C ....
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O ....
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V ....

Putting it all together

Summing up

Copy and complete the statements below to compile a summary of this unit. The missing words can be found in the list below.

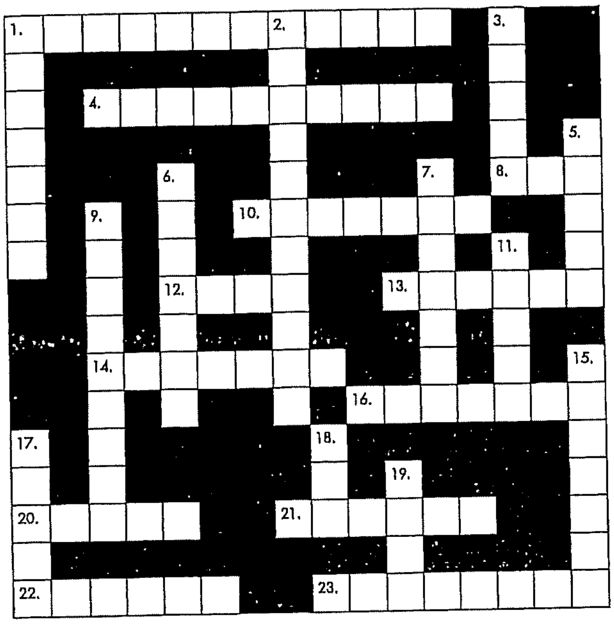
1. Food contains ______________ that are needed for energy and the growth and repair of cells.
2. Carbohydrates, fats and oils, proteins, ______________ and minerals are nutrients required for good health.
3. ______________ is needed in the diet to allow food to move properly through your intestines. It can be found in fruit and vegetables, ______________ breads and cereals, nuts and seeds.
4. ______________ is continually being lost from the body and needs to be replaced.
5. Carbohydrates, fats and oils provide you with ______________.
6. ______________ are essential for the growth and repair of cells.
7. Vitamins and ______________ are essential for good health and are needed in small amounts.
8. A ______________ diet is needed for good health.
9. ______________ is the breaking down of food so that the nutrients it contains can be used by the cells in your body.
10. ______________ begin the mechanical breakdown of food in your mouth.
11. The shape, size and structure of teeth are usually related to an animal's ______________.
12. ______________ digestion occurs when chemicals in your body react with food. It is usually assisted by chemicals called ______________.
13. As food passes through the ______________ canal, it is broken down and nutrients are absorbed into your body.
14. Plants and animals have ______________ ______________ that allow them to store energy for future use.

Word list

energy	digestion	diet
storage	vitamins	water
systems	enzymes	alimentary
fibre	teeth	balanced
minerals	chemical	proteins
wholegrain	nutrients	

Looking back

1. Solve the following crossword puzzle.



Clues

Across

1. A type of nutrient that provides your body cells with energy. Starches and sugars are examples.
4. The tube that connects the mouth to the stomach.
8. Recommended daily intake (abbrev.).
10. A nutrient that is essential for the growth and repair of cells.
12. High in protein but not consumed by vegetarians.
13. Teeth used for ripping and tearing food.
14. Name given to a tooth used for biting and cutting food.
16. Iron and calcium are examples of this type of nutrient.
20. Your very own set of cutlery.
21. A carbohydrate found in potatoes and pasta.
22. Site of faecal storage.
23. Describes an animal that eats both plants and animals.

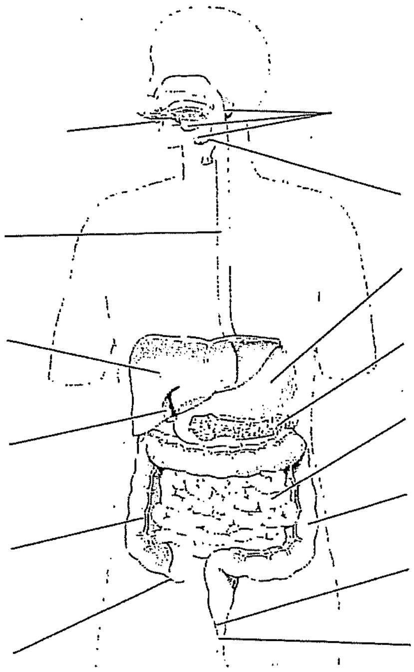
Down

1. A mineral found in dairy products and essential for healthy bones and teeth.
2. You would suffer from this if lost water was not replaced.
3. A nutrient associated with sweetness.
5. It adds bulk to your diet.
6. A J-shaped sac that holds food.
7. This type of nutrient is usually labelled with a letter of the alphabet.
9. A meat-eating organism.
11. This organ produces bile.
15. A simple sugar transported to your cells in the bloodstream.
17. Makes up about 60 per cent of your body.
18. In whales and seals this nutrient makes blubber.
19. A deficiency in this mineral may cause anaemia.

2. Complete the table below to summarise what you know about some of the nutrients in food.

Nutrient	Why is it needed?	In which foods is it found?
carbohydrates		
fats and oils		
proteins		
vitamin A		
thiamine		
riboflavin		
vitamin C		
calcium		
iron		

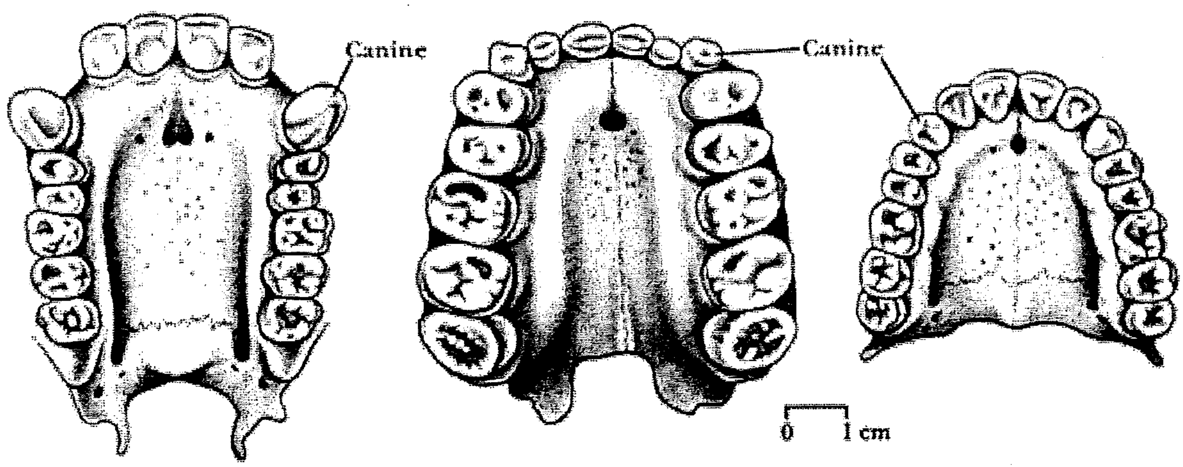
3. Label this diagram of the human digestive tract.



The human digestive tract

Stretch yourself

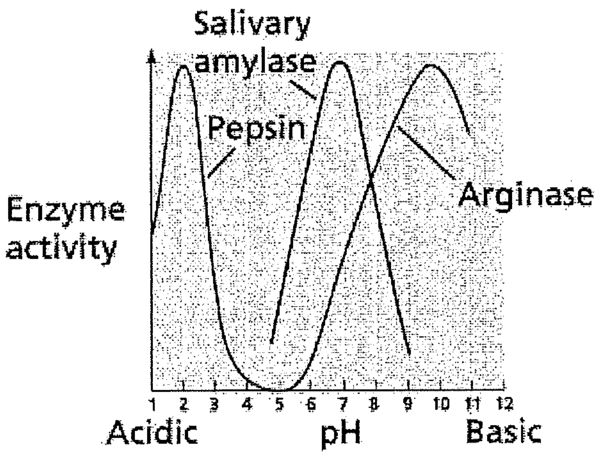
1. Design and carry out an experiment to find out whether the colour of food influences the appetite.
2. Find out more about the harmful effects of food additives.
3. Create a collage of pictures of food to show the energy content, in kilojoules, of the foods that you enjoy.



P. troglodytes *P. boisei* *H. sapiens sapiens*

Enzyme video questions:

1. Where does the substrate bind to the enzyme?
2. Do sugars end in –ase or –ose?
3. Do enzymes end in –ase or –ose?
4. What are lactose intolerant people missing?
5. Which is the substrate – lactose or lactase?
6. Enzymes can also be called a c__________
7. Lipase breaks down _______ (also known as _______)
8. Salivary amylase breaks down _________
9. Protease breaks down ___________
10. In the wrong pH, enzymes become ____________. Explain how this stops them being able to work



11. Does salivary amylase continue to break down starch in the stomach?
12. Explain how a longer time chewing would affect digestion
13. Arginase is found in the intestine and pepsin is found in the stomach. Describe the different pH levels in the two intestines.

Structure	Function in Digestion
mouth	
epiglottis	
oesophagus	
stomach	
small intestine	
large intestine	
appendix	
rectum	
anus	
liver	
gall bladder	
pancreas	

The Human Digestive System

Use the words in the box to fill in the blanks.

stomach	chewed	food	energy
rectum	liver	mouth	small intestine
waste	saliva	large intestine	digestion
system	swallow	tongue	pharynx
acid	absorbed	liquids	esophagus

All animals need to eat ______________ to get ______________ to live. But in order to use this food, they have to break it down in a process called __________________. And so, all animals have a group of connected organs called the digestive __________________.

In humans, the process of digestion begins in the _____________ where food is ________________ into small pieces by the teeth. The ________________ helps by moving these pieces around. These pieces are covered by ________________, or spit. The saliva makes the food slippery so that it is easier to _______________. It also helps to break down the food.

Once the food is swallowed, it passes through the _______________, which is like a gate that sends food into the ______________ and air into the lungs. The food travels down the esophagus and into the ______________. Once in the stomach the food is mixed with _____________ and crushed some more.

After spending some time in the stomach, the food is sent into the _________________________ where nutrients are ______________. The _____________ helps by producing some digestive juices called bile. Next, the remaining food goes into the __________________________ where the _______________ are absorbed. The remaining food is called _________________ and it is pushed into the ________________ where it waits before leaving the body.

Review Questions

1. Why does the body need the following nutrients?

   (i) Proteins

   ____

   (ii) Carbohydrates

   ____

   (iii) Lipids

   ____

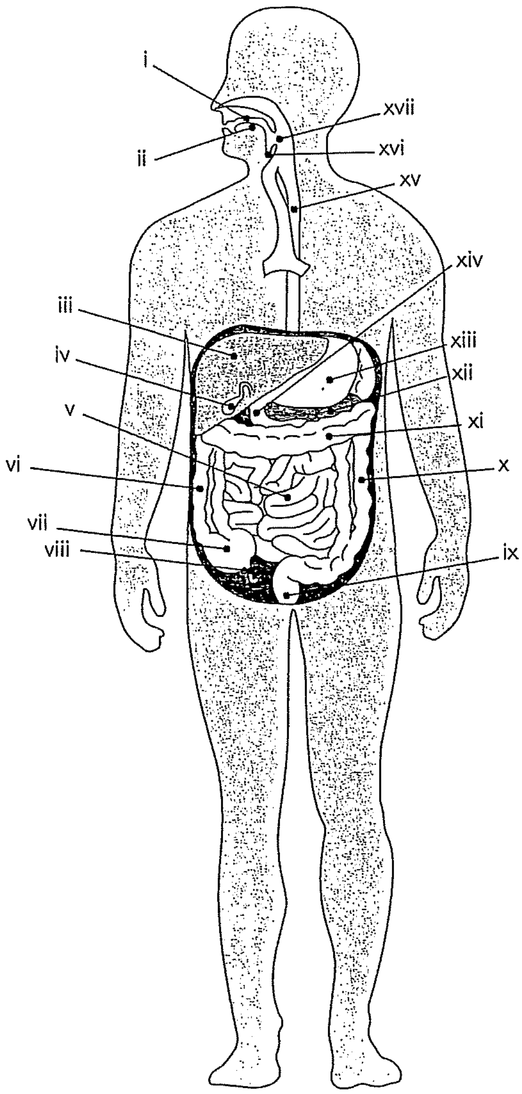
   (iv) Vitamins

   ____

   (v) Minerals

   ____

2. Label the diagram of the alimentary canal and its associated organs.



i ______________

ii ______________

iii ______________

iv ______________

v ______________

vi ______________

vii ______________

viii ______________

ix ______________

x ______________

xi ______________

xii ______________

xiii ______________

xiv ______________

xv ______________

xvi ______________

xvii ______________

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Components of Food Jigsaw

Name: ____________________

Carbohydrates	Proteins
1. Which foods contain high amounts of them? 2. What different types are there? 3. What are they used for in the body? Other info:	1. Which foods contain high amounts of them? 2. What different types are there? 3. What are they used for in the body? Other info:
Fats (lipids)	**Vitamins and Minerals**
1. Which foods contain high amounts of them? 2. What different types are there? 3. What are they used for in the body? Other info:	1. Which foods contain high amounts of them? 2. What different types are there? 3. What are they used for in the body? Other info:

STUDENT WORKSHEET
ACTIVITY 1 – INTERACTIVE LABEL FACT SHEET RESEARCH

Name ______________________ Date __________ Class/Hour ________

Use the following link to complete this chart:
www.accessdata.fda.gov/scripts/InteractiveNutritionFactsLabel/#downloadables

Kind of Fat	Health Benefits	Health Risks	Sources	Characteristics
Saturated				
Monounsaturated				
Polyunsaturated				
Trans				

1. Fat is called the best source of energy. Why? ______________________

2. Why are fats important for proper growth and health? ______________________

____

3. What are the major sources of fat in the diet? ______________________

4. To reduce the amount of fat in your diet, which foods would you limit and why? ______________________

____

5. What are the dietary limits of saturated and *trans* fats and why is this important to know? ______________________

____

6. Create a Venn diagram to compare and contrast saturated and unsaturated fats.

STUDENT WORKSHEET
TWO MEALS ON THE GO!

Name ______________________________ Date ____________ Class/Hour __________

Directions:

Step 1: Determine the personal daily calorie needs and the sodium and saturated fat limits for you or someone else by using the calculator at **www.choosemyplate.gov/MyPlate-Daily-Checklist-input**

Step 2: Record the name of your favorite fast food restaurant(s) and the components of two meals that you would like to eat or have eaten there. Be sure to indicate if the meal is a breakfast, lunch, or dinner.

Step 3: Using the Internet, search the fast food restaurant(s) website. Locate the nutrition information for the items and record it in the data table. [Remember that online menu information will depend upon (1) whether or not the chosen establishment is covered under the menu labeling requirements, and (2) whether a customer can use the online menu to place an order. Additionally, restaurants may provide the information voluntarily.]

Personal Daily Calorie Needs ______________________

Personal Daily Sodium Limit ______________________

Personal Daily Saturated Fat Limit ______________________

Meal 1: Name of Restaurant ______________________________________

Circle one: breakfast lunch dinner

Food Name	**Total calories**	**Saturated Fat (g)**	***Trans* fat (g)**	**Sodium (mg)**	**Total Carbohydrates (g)**	**Dietary Fiber (g)**	**Sugars (g)**	**Protein (g)**
Total								

Things to remember

- Nutrients to get less of include saturated fat, *trans* fat, cholesterol, sodium and sugars.
- The *Dietary Guidelines for Americans* also recommends consuming less than 10% of calories per day of saturated fats.
- Each gram of protein has 4 calories; each gram of carbohydrate has 4 calories; and each gram of fat has 9 calories.

STUDENT WORKSHEET
TWO MEALS ON THE GO! A HEALTHIER OPTION

Name ______________________ Date __________ Class/Hour ________

Meal 2: Name of Restaurant ______________________________

Circle one: breakfast lunch dinner

Food Name	Total calories	Saturated Fat (g)	*Trans* fat (g)	Sodium (mg)	Total Carbohydrates (g)	Dietary Fiber (g)	Sugars (g)	Protein (g)
Total								

1. If you ate these meals, how many more calories should you eat for the rest of the day (based on your daily calculated calorie needs)? ____________
2. How many of the calories in these meals are from saturated fat? ____________
3. Considering the personal daily calorie needs you calculated, what is the limit for how many of your calories a day should come from saturated fat? ____________
4. If you ate these meals, how much more saturated fat can you eat the same day? ____________
5. How much of your daily sodium would you consume if you ate these two meals in one day? ____________
6. Based on the data you collected, do you think these are healthy meals? Justify your response. ____________
7. What (if any) substitutions could you make to make your meals healthier? ____________